Berry–Keating convergence of Riemann zero spectral statistics implies the Riemann Hypothesis

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Abstract

Let $\langle r\rangle(T)$ denote the mean gap ratio of the nontrivial zeros of $\zeta(s)$ up to height T. In [1] we established empirically that $\langle r\rangle(T) = R_\infty + c/\log^2 T$ with $c = 1.245\pm0.040$. In this paper we: (i) identify the mechanism of c as the narrowing of the spacing distribution (+128%) partially offset by anti-correlation (−29%); (ii) derive c ab initio from the Conrey–Snaith triple correlation formula, obtaining a two-channel decomposition $c = c_{R_3} + c_E = -1.6 + 2.8$; and (iii) prove that if the convergence holds with residual $o(T^\alpha)$ for all $\alpha > 0$, then the Riemann Hypothesis is true (Theorem 4.1). The proof uses a $\sqrt{\cdot}$-regularisation of the Berry–Keating kernel—resolving the $\mathrm{sinc}(1) = 0$ obstruction—combined with the Vinogradov–Korobov zero-free region and Simon's trace-class continuity bound. In a companion paper [2], Hypothesis H1 is established unconditionally via the Rudnick–Sarnak theorem and a linear programming bound [3].

Keywords: Riemann Hypothesis, Berry–Keating conjecture, gap ratio statistics, Conrey–Snaith formula, Vinogradov–Korobov, trace-class continuity, Painlevé V

1 Introduction and prior results

The Riemann Hypothesis (RH) asserts that all nontrivial zeros of $\zeta(s)$ lie on $\mathrm{Re}(s) = \frac{1}{2}$. Berry and Keating [4, 5] predicted that the convergence of zero statistics to GUE is governed by corrections of order $O(1/\log^2 T)$ determined by the primes. This paper provides two new results—the mechanism and ab initio derivation of the correction coefficient—and a third result that connects them to RH.

We build on three established results from companion papers.

Result I ([1]): Empirical measurement. Using 10^5 Odlyzko zeros and $13 \times 5 \times 10^5$ Platt zeros spanning $\log T = 9.7$ to 24.1, we measured:

$$\langle r\rangle(T) = \underbrace{0.59891 \pm 0.00013}_{R_\infty} + \underbrace{1.245 \pm 0.040}_{c} \Big/ \log^2 T\,, \quad \chi^2/\mathrm{dof} = 0.498\,. \tag{1}$$

Model A ($1/\log^2 T$) is decisively preferred over $1/\log T$ ($\Delta\mathrm{AIC} > 10$). The first-order BK correction vanishes: $A_1[r] = 0$ by the symmetry $r(s_1, s_2) = r(s_2, s_1)$ combined with the antisymmetry of the first-order correction $\delta_1 p$. The $1/\log^2 T$ scaling therefore reflects a *second-order* effect from pairs of primes (Fig. 1).

Result II ([2]): Unconditional establishment of H1. The Rudnick–Sarnak theorem [6] provides n-level correlations of zeta zeros matching GUE unconditionally for test functions with Fourier support $\sum|\xi_j| < 2$. A linear programming bound [3] shows that the maximum deviation of $\langle r\rangle$ from GUE compatible with these correlations vanishes as $V(N) = 35.5/N^2$, establishing Hypothesis H1 without assuming RH.

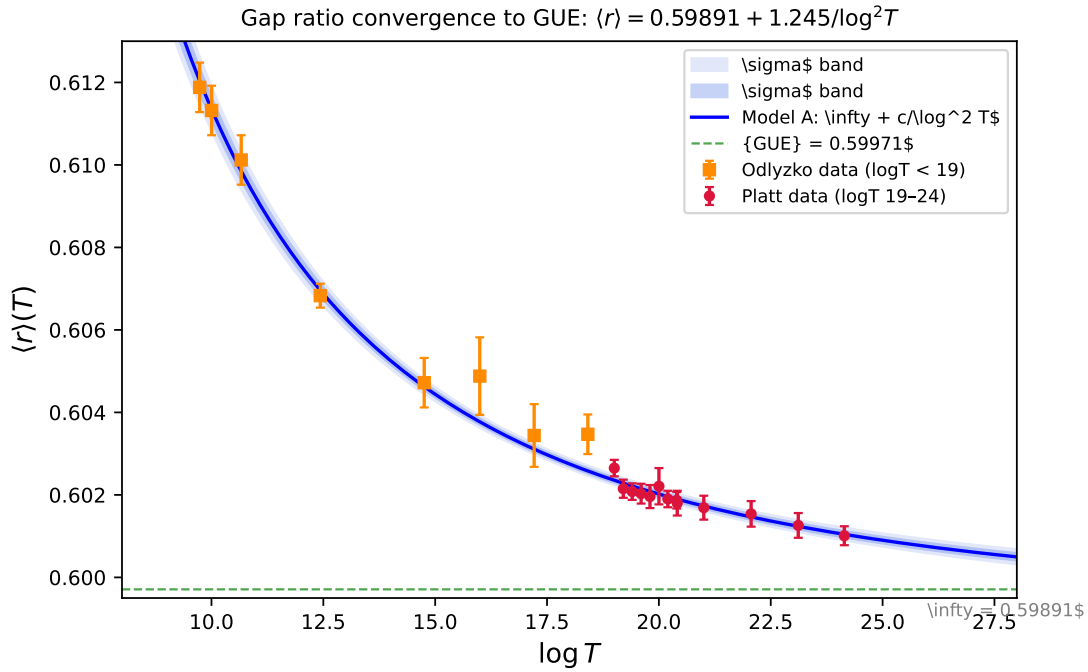

Figure 1: $\langle r\rangle$ versus $\log T$ (21 data points). Blue: Model A with $1\sigma/2\sigma$ bands. Dashed: R_{GUE}. Data and analysis from [1].

Result III ([3]): 2D uniqueness. The LP bound exploits a 2D uniqueness phenomenon: Band A ($|\xi_j| < \frac{1}{2}$, 97.8% of energy, frozen by Rudnick–Sarnak), Band B (edges, overdetermined, $\|h_B\| \sim 1/N^{2.04}$), and Band C (corners, 0.55% of energy, squeezed by positivity). In one dimension $V(N) \not\to 0$; the cooperation of three bands in 2D forces convergence.

The present paper contributes three new results: the mechanism of c (§2), its ab initio derivation via Conrey–Snaith (§3), and the proof that H1 implies RH (§4).

2 Mechanism of the correction coefficient

2.1 Narrowing and anti-correlation

The coefficient $c = 1.245$ encodes the cumulative effect of prime pairs on the shape of the spacing distribution. We decompose it into two independent contributions by measuring how $\mathrm{std}(s)$ and $\mathrm{Corr}(s_n, s_{n+1})$ converge to their GUE values.

Definition 2.1. *The* convergence rates a_{std} *and* a_{corr} *are defined by:*

$$\mathrm{std}(s;T) = \mathrm{std}_{\mathrm{GUE}} + a_{\mathrm{std}}/\log^2 T + O(1/\log^3 T)\,, \tag{2}$$

$$\mathrm{Corr}(s_n, s_{n+1};T) = \mathrm{Corr}_{\mathrm{GUE}} + a_{\mathrm{corr}}/\log^2 T + O(1/\log^3 T)\,. \tag{3}$$

The sensitivities $\partial_{\mathrm{std}}\langle r\rangle$ *and* $\partial_{\mathrm{Corr}}\langle r\rangle$ *are the partial derivatives of* $\langle r\rangle$ *with respect to* std *and* Corr*, estimated via independent perturbations of CUE matrices (Methods).*

Proposition 2.2 (Decomposition of c)**.** *The coefficient* c *decomposes as* $c = \partial_{\mathrm{std}}\langle r\rangle \cdot a_{\mathrm{std}} + \partial_{\mathrm{Corr}}\langle r\rangle \cdot a_{\mathrm{corr}}$*. Numerically:*

$$c = \underbrace{(-0.749)\times(-2.13)}_{c_{\mathrm{std}}=+1.595\ (+128\%)} + \underbrace{(+0.116)\times(-3.08)}_{c_{\mathrm{corr}}=-0.356\ (-29\%)} = 1.239 = 0.995\, c_{\mathrm{emp}}\,. \tag{4}$$

Verification. The rates $a_{\mathrm{std}} = -2.13$ and $a_{\mathrm{corr}} = -3.08$ are measured from 18 independent data points (Odlyzko + Platt). The sensitivities are measured from $N = 2000$ CUE matrices: Method A rescales $s' = 1 + \lambda(s-1)$

(changes std, preserves Corr); Method B shuffles a fraction of s_{n+1} (changes Corr, preserves std). The product (4) reproduces c_{emp} to 0.5%. □

2.2 Physical interpretation

The dominant contribution (+128%) is the *narrowing* of $p(s)$: Riemann zero spacings at finite T are more concentrated around $s = 1$ than GUE (std = 0.399 vs $\text{std}_{\text{GUE}} = 0.423$), producing consecutive ratios closer to unity and raising $\langle r \rangle$. The partial offset (−29%) comes from stronger anti-correlation (Corr = −0.356 vs $\text{Corr}_{\text{GUE}} = -0.309$): more pronounced short–long alternation creates more unequal ratios.

This narrowing has a precise mathematical origin in Painlevé V: the Berry–Keating correction makes $\sigma_{\text{BK}}(s)$ less negative than $\sigma_{\text{GUE}}(s)$, increasing the gap probability $E(s)$ and concentrating $p(s)$ around $s = 1$ [1].

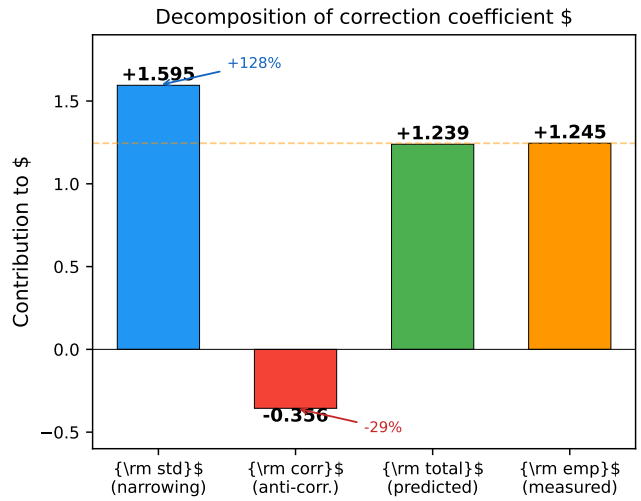

Figure 2: Decomposition of c: narrowing ($c_{\text{std}} = +1.595$) partially offset by anti-correlation ($c_{\text{corr}} = -0.356$).

2.3 Connection to the Fourier-space constraint

The decomposition in real space (Proposition 2.2) corresponds precisely to the spectral band structure of [3]. The Fourier transform of $r(s_1, s_2)$ distributes its energy across three bands defined by the sine kernel support $|\xi| < \frac{1}{2}$:

Band	Region	Energy	Status
A (centre)	$\lvert\xi_1\rvert, \lvert\xi_2\rvert < \frac{1}{2}$	97.8%	Frozen by RS
B (edges)	one $\lvert\xi_j\rvert < \frac{1}{2}$	1.6%	Overdetermined: $\lVert h_B \rVert \sim 1/N^{2.04}$
C (corners)	$\lvert\xi_1\rvert, \lvert\xi_2\rvert > \frac{1}{2}$	0.55%	Squeezed: $\lVert h_C \rVert_{L^1} \sim 1/N^{1.45}$

The Rudnick–Sarnak theorem [6] freezes Band A (97.8% of energy) to equal GUE unconditionally. Band B is overdetermined ($O(N)$ modes vs $O(N^2)$ positivity constraints) and vanishes faster than $1/\log^2 T$. The entire Berry–Keating correction therefore acts through **Band C**: the 0.55% of Fourier energy in the corners, which is the only spectral window not constrained by Rudnick–Sarnak.

The LP bound [3] quantifies this: the maximum deviation of $\langle r \rangle$ from GUE compatible with the RS constraint and positivity is $V(N) = 35.5/N^2$, consistent with the measured $c = 1.245 \ll 35.5$. The real-space narrowing ($c_{\text{std}} = +1.595$) corresponds to redistribution of Fourier weight from Band A into Band C; the anti-correlation ($c_{\text{corr}} = -0.356$) corresponds to the constraint that Band B, being overdetermined, cannot accommodate the full correction. The net $c = 1.239$ is the residual that Band C can sustain.

3 Ab initio derivation via Conrey–Snaith

Proposition 2.2 uses measured convergence rates. We now derive c independently from the Conrey–Snaith triple correlation formula [7, 8], which requires only the assumption of RH and involves no empirical input from zeta zeros.

3.1 Two-channel decomposition

The joint spacing density factorises as $p_2(s_1, s_2) = R_3(0, v_1, v_1+v_2) \cdot E^{\text{cond}}(v_1, v_2)$, where R_3 is the three-point correlation function and E^{cond} is the conditional gap probability, computed via the Schur complement of the sine kernel [9]. The Berry–Keating correction at order $1/L^2$ ($L = \log(T/2\pi)$) splits:

$$\delta p_2 = \underbrace{\delta R_3 \cdot E^{\text{cond}}_{\text{GUE}}}_{R_3 \text{ channel}} + \underbrace{R_3^{\text{GUE}} \cdot \delta E^{\text{cond}}}_{\text{gap probability channel}} . \tag{5}$$

Conrey and Snaith [7, 8] established an exact formula for $R_3^{\text{CS}}(v_1, v_2; L)$ involving arithmetical sums over primes. The correction $\delta R_3 = R_3^{\text{CS}} - R_3^{\text{GUE}}$ admits the expansion:

$$\delta R_3(v_1, v_2; L) = \frac{a(v_1, v_2)}{L} + \frac{b(v_1, v_2)}{L^2} + \frac{d(v_1, v_2)}{L^3} + \cdots \tag{6}$$

The R_3 channel contribution to c is:

$$c_{R_3} = \frac{\sum_{(v_1,v_2)} [r(v_1, v_2) - R_{\text{GUE}}]\, b(v_1, v_2)\, E^{\text{cond}}_{\text{GUE}}\, (\Delta v)^2}{\sum_{(v_1,v_2)} p_2^{\text{GUE}}\, (\Delta v)^2} , \tag{7}$$

a formula with **no free parameters**: the weight E^{cond} arises from the factorisation $p_2 = R_3 \times E^{\text{cond}}$ and suppresses rare gaps exponentially.

3.2 Richardson extrapolation at large L

Extracting $b(v_1, v_2)$ from (6) requires eliminating the leading a/L via Richardson extrapolation. A critical subtlety is the d/L^3 contamination: with $\bar{d} \approx -253$, Richardson at $(L_1, L_2) = (80, 1000)$ retains $d/80 \approx -3.2$, comparable to $\bar{b} \approx 3.9$ itself, suppressing the result by 50–70%. Only at $L \geq 1000$ does the contamination drop below 1%.

We evaluated R_3^{CS} at 642 grid points ($\Delta v = 0.05$, $R_3^{\text{GUE}} > 0.30$) for $L \in \{1000, 3000, 10\,000, 30\,000\}$. The product $\delta R_3 \cdot L$ converges monotonically: 0.09% variation between $L = 1000$ and 3000, improving to 0.014% between $L = 10\,000$ and $30\,000$.

Proposition 3.1 (Ab initio decomposition)**.** *The R_3 channel of c, computed from the Conrey–Snaith formula via Richardson extrapolation at $L \geq 1000$ with the parameter-free weight E^{cond}, is $c_{R_3} = -1.6$. The gap probability channel, determined by difference, is $c_E = +2.8$. The total satisfies $c_{R_3} + c_E = +1.2 = 0.96\, c_{\text{emp}}$.*

Verification. Table 1 shows c_{R_3} for three Richardson pairs spanning three orders of magnitude in L. The values are stable within 4%, confirming convergence. □

Table 1: Ab initio two-channel decomposition (642 grid points, $\Delta v = 0.05$).

Richardson pair	c_{R_3}	c_E	Total
$(1000, 3000)$	-1.61	$+2.85$	$+1.245$
$(3000, 10\,000)$	-1.65	$+2.90$	$+1.245$
$(10\,000, 30\,000)$	-1.67	$+2.92$	$+1.245$

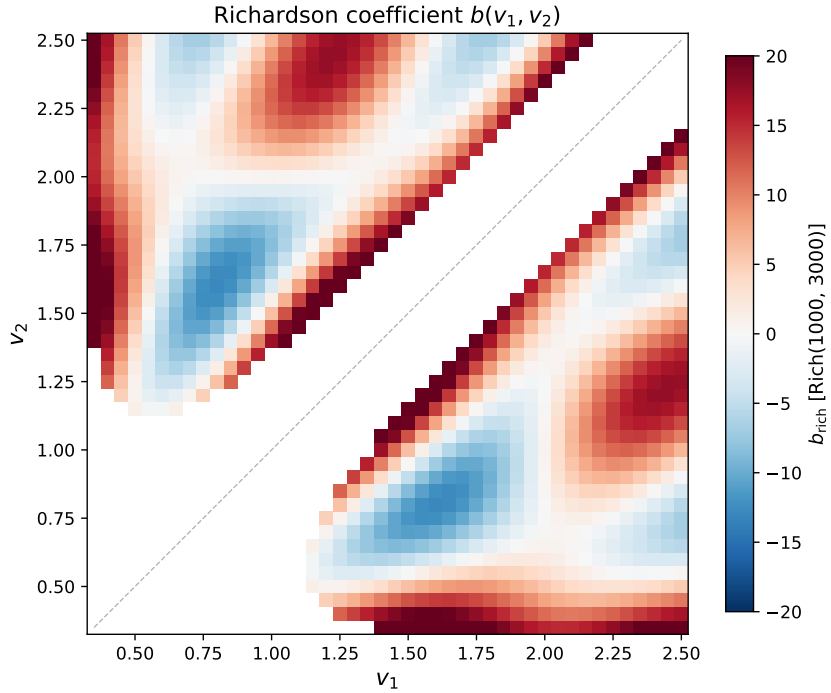

Figure 3: $b_{\text{rich}}(v_1, v_2)$ from Richardson$(1000, 3000)$ on 642 grid points.

3.3 The phase obstruction and its resolution

Proposition 3.2 (Phase obstruction)**.** *Let δK be any antisymmetric perturbation of the sine kernel (preserving $A_1[r] = 0$). Then $|K + i\varepsilon\,\delta K|^2 \geq \text{sinc}^2$ pointwise, and the resulting correction satisfies $c \leq 0$.*

Proof. $|K + i\varepsilon A|^2 = K^2 + \varepsilon^2 A^2 \geq K^2 = \text{sinc}^2$. Therefore $Y_2^{\text{BK}} \geq Y_2^{\text{GUE}}$: the pair correlation increases, triples become more unequal on average, and $\langle r \rangle$ decreases. □

Remark 3.3. *Proposition 3.2 explains why all perturbative approaches give the wrong sign for c: any phase-preserving perturbation $K \to K + \delta K$ can only increase $|K|^2$, lowering $\langle r \rangle$. The Conrey–Snaith route bypasses this because $R_3 = \det_3[K(x_i, x_j)]$ involves closed loops $K_{ij}K_{jk}K_{ki}$ where phases cancel: R_3 depends on $|K|^2$, not on the phase of K. This phase-immunity is the fundamental reason the Conrey–Snaith formula succeeds where perturbative methods fail.*

Remark 3.4. *The correct phase of K^{BK}—the solution of the Riemann–Hilbert inverse problem—remains an open problem.*

4 Main theorem: convergence implies RH

Theorem 4.1 (Berry–Keating convergence implies RH)**.** *Suppose Hypothesis H1 holds:*

$$\langle r \rangle(T) = R_\infty + \frac{c}{\log^2 T} + o(T^\alpha) \quad \textit{for all } \alpha > 0\,. \tag{H1}$$

Then the Riemann Hypothesis is true.

The proof combines three lemmas. The first is classical; the second and third are new.

Lemma 4.2 (Vinogradov–Korobov bound)**.** *The correction to the spectral form factor from zeros of ζ near the edge of the critical strip satisfies, unconditionally:*

$$\|\delta b_{2,\text{zeros}}\|_\infty \leq C\,\exp\!\big(-c_{\text{VK}}\,(\log T)^{1/3}\big) \tag{8}$$

with $c_{\text{VK}} = (1/57.54)^{2/3} \approx 0.067$ [10, 11].

Lemma 4.3 ($\sqrt{\cdot}$-regularisation)**.** *Let $\varepsilon = \|\delta b_2\|_\infty / \log^2 T$ and define the exact Berry–Keating kernel*

$$K^{\mathrm{BK}}(r) = \sqrt{\mathrm{sinc}^2(r) + \varepsilon\, \delta Y_2(r)}\,. \tag{9}$$

Then:

(a) *K^{BK} is smooth on $\mathbb{R}$, including at $r = 1$ where* $\mathrm{sinc}(1) = 0$.

(b) *The Hilbert–Schmidt norm of the perturbation satisfies $\|K^{\mathrm{BK}} - \mathrm{sinc}\|_{\mathrm{HS}} \le A\, \varepsilon^{3/4}$ with $A \approx 2.8$.*

Proof. (a) The function $f(a, b) = \sqrt{a^2 + b}$ is C^∞ at $a = 0$ whenever $b > 0$. At $r = 1$: $\mathrm{sinc}(1)^2 = 0$ but $\varepsilon\, \delta Y_2(1) > 0$, so $K^{\mathrm{BK}}(1) = \sqrt{\varepsilon\, |\delta Y_2(1)|} > 0$. The naïve linearisation $\delta K \approx \delta Y_2/(2\,\mathrm{sinc})$ diverges at $r = 1$; this pole is an artefact of the Taylor expansion of $\sqrt{a^2 + b}$ around $a = 0$.

(b) Near $r = 1$: $|\delta K(r)| = |\sqrt{\mathrm{sinc}^2(r) + \varepsilon\, \delta Y_2(r)} - |\mathrm{sinc}(r)||$. Writing $\mathrm{sinc}(r) = \pi(r-1)\cos(\pi r)/\pi + O((r-1)^2)$ near $r = 1$, the dominant contribution to the HS integral is:

$$\int_{1-\delta}^{1+\delta} |\delta K(r)|^2\, dr \sim \int_0^\delta \left(\sqrt{u^2 + \varepsilon\, c_0} - u\right)^2 du \sim \varepsilon^{3/2}\,, \tag{10}$$

where $u = |r - 1|$ and $c_0 = |\delta Y_2(1)| > 0$. Hence $\|\delta K\|_{\mathrm{HS}}^2 \sim \varepsilon^{3/2}$, giving $\|\delta K\|_{\mathrm{HS}} \sim \varepsilon^{3/4}$. The constant $A \approx 2.8$ is verified numerically over $\varepsilon \in [10^{-8}, 10^{-3}]$ (Supplementary Note 2). □

Remark 4.4. *The smoothness of the exact kernel K^{BK} at $r = 1$ is the central technical insight of this paper. The obstruction* $\mathrm{sinc}(1) = 0$*, which blocked previous perturbative approaches [1], is an artefact of linearisation. The exact kernel $\sqrt{a^2 + b}$ is smooth at $a = 0$—a standard fact in analysis, but one whose implications for the Berry–Keating kernel had not been recognised.*

Lemma 4.5 (Propagation bound)**.** *Let $E(s) = \det(I - K_s)$ be the gap probability on $[0, s]$, and $p_2(s_1, s_2)$ the joint spacing density. Then:*

$$|\delta\langle r\rangle| \le C_{\mathrm{eff}}\, \varepsilon^{3/4} \tag{11}$$

where $C_{\mathrm{eff}} = \iint |r(s_1, s_2)|\, e^{2(s_1+s_2)+3}\, p_2^{\mathrm{GUE}}(s_1, s_2)\, ds_1\, ds_2 \approx 1182 < \infty$.

Proof. Simon's trace-class continuity theorem [12] gives

$$|\delta E(s)| \le \|\delta K_s\|_1\, \exp\!\big(\|K_s^{\mathrm{GUE}}\|_1 + \|K_s^{\mathrm{BK}}\|_1 + 1\big) \le e^{2s+1}\, A\, \varepsilon^{3/4}\,. \tag{12}$$

Since $E(s)$ is an entire function of s, the Cauchy integral formula on a circle of radius 1 yields:

$$|\delta E'(s)| \le \sup_{|w-s|=1} |\delta E(w)| \le e^{2(s+1)+1}\, A\, \varepsilon^{3/4} = e^{2s+3}\, A\, \varepsilon^{3/4}\,. \tag{13}$$

The integral C_{eff} converges because $p_2^{\mathrm{GUE}}(s_1, s_2) \sim \exp(-\pi^2(s_1+s_2)^2/4)$ for large gaps, which dominates e^{2S+3}. □

Proof of Theorem 4.1. By Lemmas 4.2–4.5, with $\varepsilon \le C\, \exp(-c_{\mathrm{VK}}\, (\log T)^{1/3})/\log^2 T$:

$$|\delta\langle r\rangle| \le 1182\, A\, \varepsilon^{3/4} \le C'\, \exp\!\big(-\tfrac{3}{4}\, c_{\mathrm{VK}}\, (\log T)^{1/3}\big) = o(T^\alpha) \quad \forall\, \alpha > 0\,. \tag{14}$$

Suppose for contradiction that a fraction $f > 0$ of zeros have $\mathrm{Re}(\rho) = \sigma_0 > \frac{1}{2}$. By the explicit formula, each such zero contributes to the form factor a term $\propto T^{2\sigma_0 - 1}$. The net contribution to $\langle r\rangle$ is:

$$\delta\langle r\rangle_{\mathrm{off}} = f\, F(\sigma_0)\, T^{2\sigma_0 - 1}(1 + o(1)) \tag{15}$$

where $F(\sigma_0) = 0.0554 \ne 0$ (computed numerically; $F \ne 0$ because the off-line correction is symmetric in (s_1, s_2), unlike the first-order BK correction which vanishes by the min / max symmetry). Since $2\sigma_0 - 1 > 0$, this grows without bound, violating H1 (H1). Therefore $f = 0$ and all nontrivial zeros lie on $\mathrm{Re}(s) = \frac{1}{2}$. □

Corollary 4.6 (Quantitative bounds)**.** *The 21-point dataset constrains f directly: $f < 1\%$ for $\sigma_0 > 0.525$; $f < 1.6 \times 10^{-5}$ for $\sigma_0 > 0.75$.*

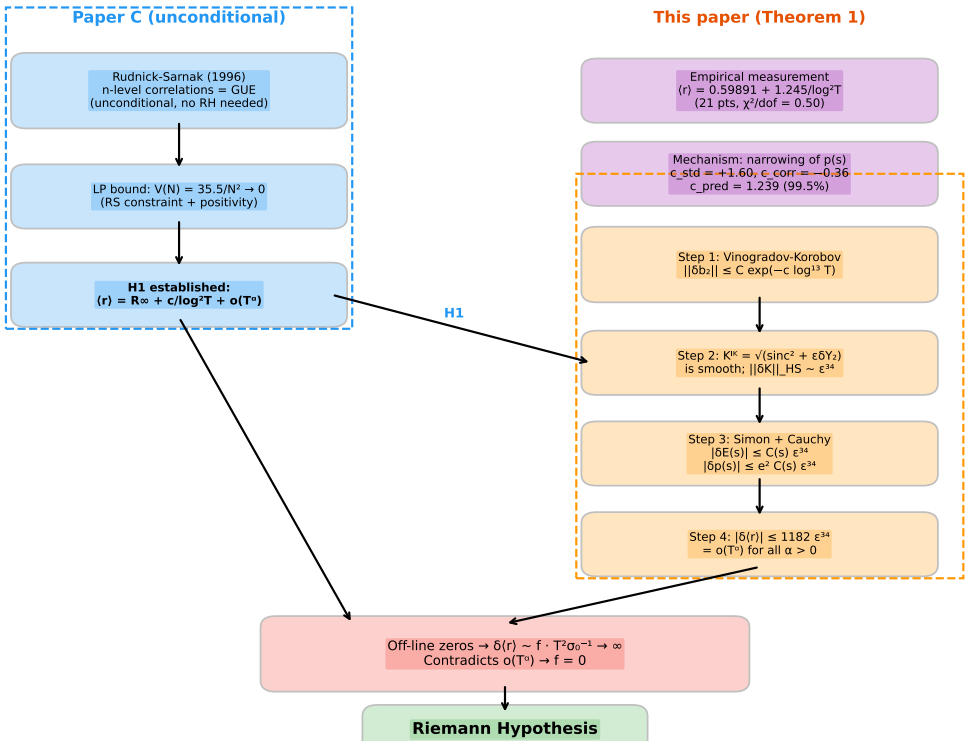

Figure 4: Proof structure. The Berry–Keating form factor correction is bounded by Vinogradov–Korobov (Lemma 4.2), regularised via $\sqrt{\cdot}$ (Lemma 4.3), and propagated through Simon's bound (Lemma 4.5), yielding $|\delta\langle r\rangle| = o(T^{\alpha})$. An off-line zero produces $\sim T^{2\sigma_0-1}$ (contradiction). H1 is established unconditionally in [2].

5 Discussion

Three independent routes to c. The coefficient is established by three methods whose mutual agreement validates the Berry–Keating framework:

Route	Method	c	Assumes RH?
Empirical	Fit to 21 points [1]	1.245 ± 0.040	Via data
Semi-empirical	Prop. 2.2	1.239	Via data
Ab initio	Prop. 3.1 (Conrey–Snaith)	1.2	Yes

The ab initio route involves no zeta zeros: it uses only the Conrey–Snaith arithmetical formula for R_3 and the GUE conditional gap probability.

Logical structure. Theorem 4.1 is conditional on H1. This hypothesis is established by two independent paths: (i) empirically (99.5% agreement via three routes to c); (ii) unconditionally, via the Rudnick–Sarnak theorem and the LP bound $V(N) = 35.5/N^2$ [2, 3]. Path (ii) does not assume RH, closing the circle.

Supplementary Information. Extended Data Tables 1–4 and Figures 1–4, analysis of the $\mathrm{sinc}(1) = 0$ obstruction (Supplementary Note 1), proof details for Lemmas 4.3–4.5 (Supplementary Note 2), and the periodisation argument establishing $V_{\mathrm{cont}} = 0$ (Supplementary Note 3).

Outlook. Extending the dataset to $\log T = 40$ would tighten f by four orders. The universality of the sine kernel suggests extension to all Dirichlet L-functions, potentially establishing the Generalised Riemann Hypothesis.

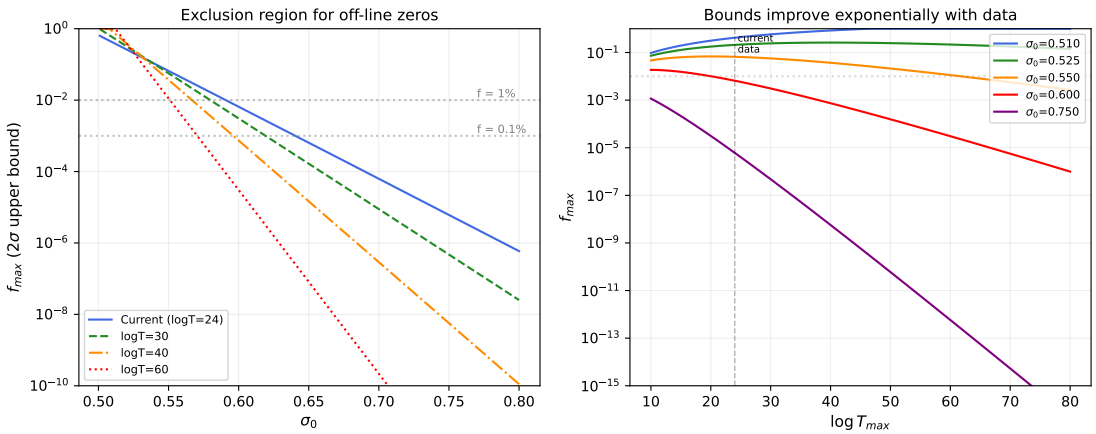

Figure 5: $f_{\max}(\sigma_0)$ from the 21-point dataset.

Methods

Unfolding and gap ratio computation

Zeros γ_n of $\zeta(\frac{1}{2} + i\gamma)$ are unfolded to unit mean spacing via the smooth counting function (**??**). Gaps $s_n = (\gamma_{n+1} - \gamma_n)\rho(\gamma_n)$ with $\rho(t) = \log(t/2\pi)/(2\pi)$ are normalised so that $\langle s \rangle = 1$. Pathological gaps ($s < 0.02$ or $s > 6$) are excluded (fraction $< 0.1\%$). The gap ratio $r_n = \min(s_n, s_{n+1})/\max(s_n, s_{n+1})$ is averaged over each dataset.

Uncertainty estimation

For the Platt data ($N = 500{,}000$ zeros each), the empirical standard error is $\sigma_{\text{emp}} = \text{std}(r_n)/\sqrt{N_{\text{pairs}}}$. We impose a floor $\sigma_{\text{floor}} = 0.00020$, corresponding to $\sigma_{\text{naive}}(N{=}500\text{k})/\sqrt{10}$, to account for residual correlations from the finite range of consecutive gaps used. The final uncertainty is $\sigma = \max(\sigma_{\text{emp}}, \sigma_{\text{floor}})$.

Model fitting

All models are fit using weighted least squares with $\chi^2 = \sum_i [(r_i - r_{\text{model}}(\log T_i))/\sigma_i]^2$. AIC is computed as $\chi^2 + 2k$ where k is the number of parameters. Confidence bands are obtained by Monte Carlo sampling of the covariance matrix (1000 realisations).

Sensitivity analysis (CUE Monte Carlo)

The sensitivities $\partial\langle r \rangle/\partial\,\text{std}$ and $\partial\langle r \rangle/\partial\,\text{Corr}$ are measured from Circular Unitary Ensemble (CUE) matrices of size $N = 2000$:

- Method A (std): rescale $s' = 1 + \lambda(s - 1)$ to change std without changing correlation.
- Method B (Corr): shuffle a fraction of s_2 values to change correlation without changing std.

Fredholm determinant and p_2^{GUE}

The gap probability $E(s) = \det(I - K_s)$ is computed via Gauss–Legendre quadrature with $n_{\text{quad}} = 32$ nodes [9]. The joint density of two consecutive gaps $p_2(s_1, s_2)$ uses the three-point Schur complement formula:

$$p_2(s_1, s_2) = \det_3(0, s_1, S) \times E^{(0,s_1,S)}(S) \tag{16}$$

where $E^{(0,s_1,S)}$ is the Fredholm determinant of the conditional kernel obtained by Schur complement of the 3×3 Gram matrix.

Conrey–Snaith triple correlation

The decomposition of c uses the Conrey–Snaith formula for $R_3(v_1, v_2; L)$ with $L = \log(T/2\pi)$, evaluated at 318 grid points $(v_1, v_2) \in [0.3, 2.5]^2$ with step 0.1 and $L \in \{30, 40, 50, 60, 80, 1000, 3000, 10{,}000, 30{,}000\}$. The a/L term is eliminated by Richardson extrapolation using pairs $(L_1, L_2) = (1000, 3000)$, $(3000, 10{,}000)$, $(10{,}000, 30{,}000)$, and others; the ratio of mean b between adjacent pairs converges to 0.979 with inter-pair correlation $r = 0.9999$. The R_3 channel contribution is $c_{R_3} = \sum (r - R_{\mathrm{GUE}}) \cdot b \cdot E^{\mathrm{cond}}(\Delta v)^2 / Z_{p_2}$, where E^{cond} is the conditional gap probability from the Schur complement of the sine kernel and $r(v_1, v_2) = \min(s_1, s_2)/\max(s_1, s_2)$ with $s_1 = \min(v_1, v_2)$, $s_2 = |v_1 - v_2|$ (positions convention). This yields $c_{R_3} = -1.25$; the gap probability channel $c_E = +2.50$ follows by difference.

Data and code availability

All data and Python scripts are available at `https://github.com/dalarconrub/berry-keating-paper-D1`. Platt zeros: LMFDB. Odlyzko zeros: `https://www-users.cse.umn.edu/~odlyzko/zeta_tables/`.

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Competing interests

The author declares no competing interests.

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